$$D = \begin{vmatrix} 1 & 1 \\ 1 & -2 \end{vmatrix} = [1 \times (-2)] - (1 \times 1) = -2 - 1 = -3$$

$$D_x = \begin{vmatrix} 20 & 1 \\ 2 & -2 \end{vmatrix} = [20 \times (-2)] - (1 \times 2) = -40 - 2 = -42$$

$$D_y = \begin{vmatrix} 1 & 20 \\ 1 & 2 \end{vmatrix} = (1 \times 2) - (20 \times 1) = 2 - 20 = -18$$

$$x = \frac{D_x}{D} \text{ and } y = \frac{D_y}{D}$$

$$\therefore x = \frac{-42}{-3} \text{ and } y = \frac{-18}{-3}$$

$$\therefore x = 14, \quad y = 6$$

$\therefore$ Length of the rectangle is 14 cm and breadth is 6 cm.

Ex. (2)

Sale ! Sale !! Sale !!! only for 2 days

I have some analogue wrist watches and some digital wrist watches. I am going to sell them at a discount

Sale of 1st day
Analogue watch = 11
Digital watch = 6
Received amount = ₹ 4330

Sale of the 2nd day
Analogue watch = 22
Digital watch = 5
Received amount = ₹ 7330

Find selling price of wrist watch of each type.

Solution : Let selling price of each analogue watch be ₹ x

Selling price of each digital watch be ₹ y

From first condition –

$$11x + 6y = 4330 \quad \ldots \quad \text{(I)}$$

from 2nd condition –

$$22x + 5y = 7330 \quad \ldots \quad \text{(II)}$$

multiplying equation (I) by 2 we get,

$$22x + 12y = 8660 \quad \ldots \quad \text{(III)}$$

subtract equation (III) from equation (II).

$$\begin{array}{rcl} 22x + 5y &=& 7330 \quad \ldots \quad \text{(II)} \\ -\;22x \underset{-}{+} 12y &=& \underset{-}{}8660 \quad \ldots \quad \text{(III)} \\ \hline -7y &=& -1330 \end{array}$$

$$\therefore \quad y = 190$$

Substitute this value of y in equation (I)

$$11x + 6y = 4330$$

$$\therefore \quad 11x + 6(190) = 4330$$

$$\therefore \quad 11x + 1140 = 4330$$

$$\therefore \quad 11x = 3190$$

$$\therefore \quad x = 290$$

$\therefore$ selling price of each analogue watch is ₹ 290 and of each digital watch is ₹ 190.